

How Many Words Until You Know? A Complete Disambiguation Map of the Quran

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Abstract

We present the first complete, parameter-free disambiguation map of the Quran. For each of the 6,236 ayahs, at every word position, we compute the minimum number of consecutive words needed to uniquely identify that ayah via exact word-sequence matching. The method is deterministic, reproducible, and runs in 6.5 seconds on a single core.

The main findings: 94.6% of ayahs are identifiable from their opening words, requiring a mean of 3.11 words. 339 ayahs are never unique in isolation—but under continuous recitation, where windows span ayah boundaries, 328 of those 339 become identifiable, leaving only 11 truly unresolvable ayahs in the entire Quran. We release six JSON artifacts (77 MB) and all code publicly, and outline open questions we hope others will pursue.

1 Introduction

The Quran, regarded by Muslims as the verbatim word of God (*kalām Allāh*), comprises 6,236 ayahs across 114 surahs—approximately 78,000 words. A distinctive feature of this text is its extensive use of repeated phrasing: refrains, oaths, prophet-narrative structures, and closing phrases that recur across surahs. The Islamic scholarly tradition recognizes these as deliberate rhetorical emphasis (*takrār*), serving purposes of reinforcement, contemplation, and structural unity.

From a computational perspective, this repeated phrasing raises a simple question: **given a stream of words from the Quran, how many must an algorithm hear before it can uniquely identify which ayah is being recited?**

This is relevant to automatic verse identification, Quranic information retrieval, and computational stylistics. We address it exhaustively: for every ayah, from every possible starting word position, we compute the answer. The result is a complete disambiguation map—a dataset that, to our knowledge, has not previously existed.

We use the term *confuser* throughout this paper to describe ayahs whose word sequences overlap, creating ambiguity for an identification algorithm. This is a property of the algorithmic task, not an attribute of the Quranic text itself.

Our contributions:

1. A complete disambiguation map: 6,236 ayahs $\times$ all starting positions (78,092 window analyses).
2. A parameter-free, deterministic method that runs in 6.5 seconds.
3. Corpus-level findings on disambiguation patterns, repeated phrases, confuser networks, and continuous-recitation analysis.
4. Six publicly released dataset artifacts and all source code.
5. Open questions we hope this data enables others to explore.

Algorithm 1 Disambiguation for one ayah

Require: Ayah index a , word sequence $\mathbf{w} = (w_1, \dots, w_m)$, index $\mathcal{I}$

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1: for each starting position  $s \in \{0, 1, \dots, m-1\}$  do
2:   for  $\ell = 1, 2, \dots, m - s$  do
3:      $\mathbf{q} \leftarrow (w_{s+1}, \dots, w_{s+\ell})$ 
4:      $C \leftarrow \mathcal{I}[\mathbf{q}] \setminus \{a\}$  ▷ other ayahs sharing this sequence
5:     if  $|C| = 0$  then
6:        $d(a, s) \leftarrow \ell$ ; break ▷ uniquely identified
7:     end if
8:   end for
9:   if never reached  $|C|=0$  then
10:     $d(a, s) \leftarrow -1$  ▷ never unique from this position
11:   end if
12: end for
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2 Method

2.1 Text Normalization

We normalize the Uthmani script by stripping diacritical marks (*tashkīl*), collapsing *hamzah* variants to bare *alif*, normalizing *tā' marbūṭah* to *hā'* and *alif maqṣūrah* to *yā'*, removing elongation marks and punctuation, and collapsing whitespace. Each ayah is tokenized by whitespace. The normalized corpus contains 6,236 ayahs with word counts ranging from 1 to 128 (mean 12.5, median 10).

2.2 N-Gram Inverted Index

We build an inverted index mapping every contiguous word n -gram ($n = 1, \dots, 15$) to the set of ayahs containing it, yielding 525,402 unique entries. For windows longer than 15 words, we initialize candidates from the 15-gram lookup and verify the full sequence—efficient because few candidates remain at that length.

2.3 Progressive Window Disambiguation

Algorithm 1 runs for each ayah across all starting positions. The method has **zero parameters**: no similarity thresholds, no fuzzy matching, no cutoffs. Given the same text and normalization, any implementation produces identical results. The full computation completes in **6.5 seconds** on a single CPU core.

2.4 Worked Examples

To make the algorithm concrete, we trace it on three ayahs.

Algorithm 2 Trace: Q 2:138 — One word, instantly identified

ℓ	Words heard so far	Other ayahs
1	صِبْغَةً	0 ✓

The word صِبْغَةً (“the colouring [of God]”) appears in no other ayah. One word is enough.

Algorithm 3 Trace: Q 4:43 — Same ayah, two very different entry points

From the beginning:

ℓ	Words heard so far	Other ayahs
1	يَا أَيُّهَا	141
2	يَا أَيُّهَا الَّذِينَ	91
3	يَا أَيُّهَا الَّذِينَ ءَامَنُوا	88
4	يَا أَيُّهَا الَّذِينَ ءَامَنُوا لَا	26
5	يَا أَيُّهَا الَّذِينَ ءَامَنُوا لَا تَقْرَبُوا	0 ✓

From word 15 (mid-verse):

1	عَابِرِي	0 ✓
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Five words from the opening, one word from the middle. The opening phrase يَا أَيُّهَا الَّذِينَ ءَامَنُوا (“O you who believe”) begins 89 ayahs—it takes two more words to narrow down. But mid-verse, عَابِرِي (“travelers”) is unique in the entire Quran.

Algorithm 4 Trace: Q 2:14 — A persistent single confuser

From the beginning:

ℓ	Words heard so far	Other ayahs
1	وَإِذَا	127
2	وَإِذَا لَقُوا	1 Q 2:76
3	وَإِذَا لَقُوا الَّذِينَ	1 persists...
(Q 2:76 continues matching for words 4–7)		
8	... خَلَوْا	0 ✓

From the last word:

1	مُشْتَهَرُونَ	0 ✓
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After the first word, 127 candidates drop to just 1: Q 2:76, which shares the first 7 words. That single confuser holds on for 6 more words before the ayahs finally diverge at word 8. Meanwhile, from the end, مُشْتَهَرُونَ is unique in the entire Quran—one word is enough. We call Q 2:76 the *persistent confuser* of Q 2:14: the last one standing.

3 Findings

3.1 The Big Picture

Table 1 and Figure 2 present the headline results. The Quran’s normalized vocabulary contains 14,790 unique words, of which 8,674 (58.6%) are hapax legomena—words that appear exactly once in the entire text. This high proportion of unique vocabulary means that most ayahs contain at least one word found nowhere else.

Starting from the first word, 94.6% of ayahs can be uniquely identified, requiring a mean of only 3.11 words. Figure 1 shows rapid convergence: 44.3% disambiguate within 2 words, 68.6% within 3, and 89.4% within 6.

Statistic	Value
Total ayahs	6,236
Identifiable from start	5,897 (94.6%)
Never unique from start	339 (5.4%)
Mean words to identify (from start)	3.11
Median	3
Min / Max	1 / 18
Identifiable via boundary context	328 of 339
Truly unresolvable ayahs	11
Unique vocabulary (hapax legomena)	8,674 of 14,790 (58.6%)

Table 1: Corpus-level summary.

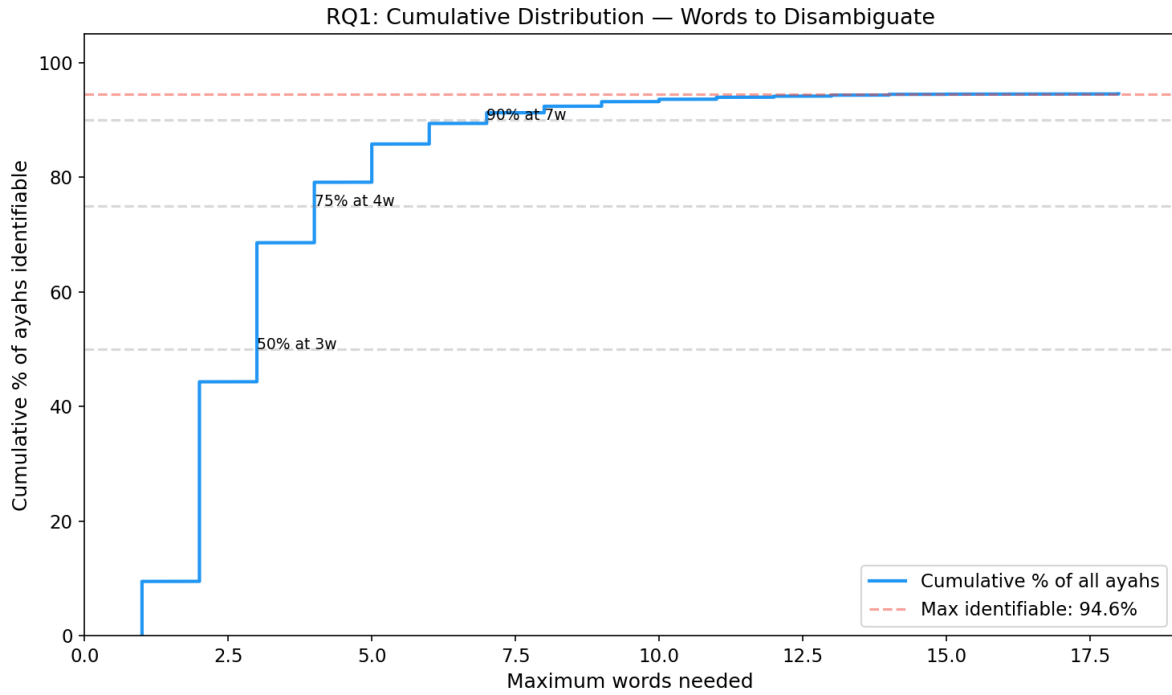


Figure 1: Cumulative distribution: how quickly ayahs become identifiable from their opening word.

592 ayahs are identifiable from their very first word. Every one of these opens with a hapax legomenon. The shortest unique openers are just two Arabic characters: سَلْ (“Ask,” Q 2:211), ذُقْ (“Taste!” Q 44:49), and فَكْ (“Freeing,” Q 90:13).

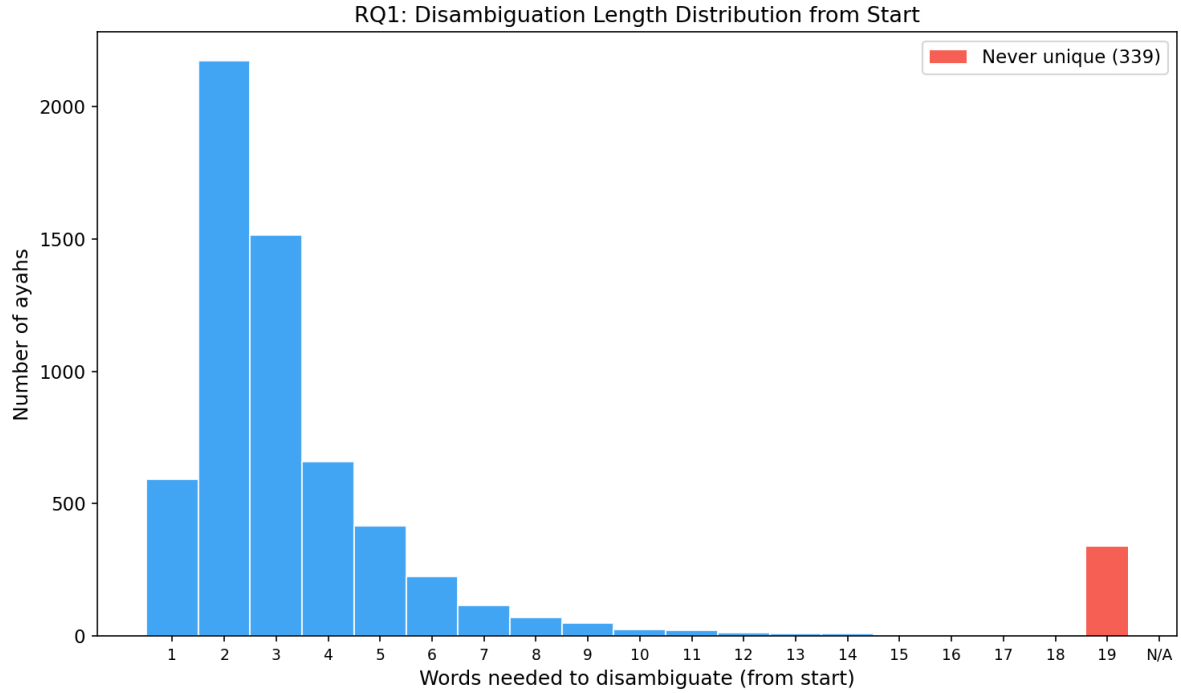


Figure 2: Distribution of disambiguation lengths from start. The 339 never-unique ayahs are shown in red.

3.2 The 339 Never-Unique Ayahs

339 ayahs cannot be uniquely identified from their start at any length, because their entire text is shared with at least one other ayah. These cluster into clear structural categories:

- **The basmalah** (Q 1:1): بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ appears verbatim at the start of 113 surahs.
- **Surat al-Rahman’s refrain** (Q 55): رَبِّكَمَا تُكَذِّبَانِ (“Which of your Lord’s favors will you deny?”) is repeated identically 31 times.
- **Surat al-Shu‘ara’s prophet-narrative phrases** (Q 26): 53 ayahs share repeated phrases across the surah’s seven parallel prophet stories.
- **Al-Saffat** (Q 37): 28 ayahs. **Al-Mursalat** (Q 77): 14 ayahs.
- **Disconnected opening letters** (*hurūf muqatta‘āt*): e.g., *Alif-Lām-Mīm* opens six surahs identically.
- **Verbatim-repeated passages**: e.g., Q 2:134 and Q 2:141 share all 15 words.

However, as we show in §3.7, 328 of these 339 become identifiable when boundary context from the surrounding ayahs is available. Only 11 remain truly unresolvable—and they are fascinating (§3.8).

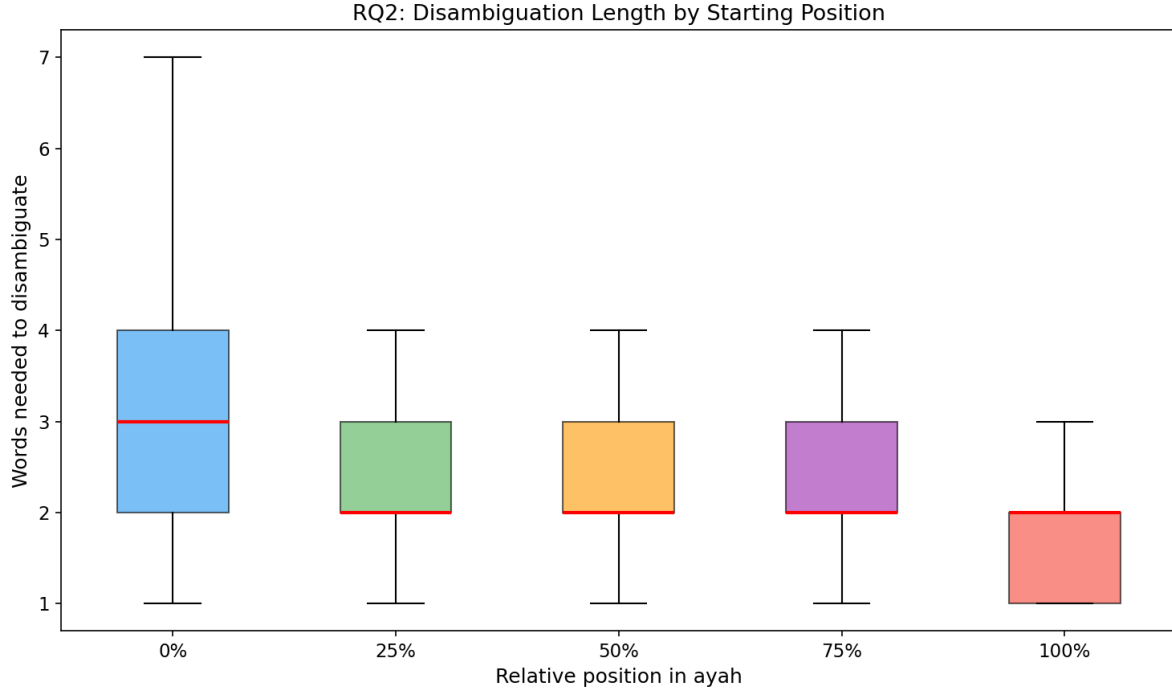


Figure 3: Disambiguation length by relative starting position. Endings are systematically easier than beginnings.

3.3 Where in the Ayah Does Identification Happen?

Figure 3 shows a clear pattern: ayah endings are easier to identify than beginnings. Mean disambiguation drops from 3.22 words at the start to 1.80 at the end. Openings often share common phrases (يَا أَيُّهَا الَّذِينَ آمَنُوا opens 89 ayahs), while closing phrases tend to be more distinctive.

An interesting corollary: for easy ayahs ($d \leq 3$), the disambiguating word falls in the first 20% of the ayah. For difficult ayahs ($d \geq 8$), it shifts to the second half (mean: 60% through the ayah). The disambiguating word marks the exact boundary between shared and unique content.

3.4 Surah-Level Patterns

Figure 4 ranks all 114 surahs. The five requiring the most words: Al-Mujadilah (4.73), Al-Taghabun (4.72), Al-‘Asr (4.67), Al-Rum (4.07), and Al-Hajj (4.05). The easiest: Al-Fatihah (1.50), Al-‘Adiyat (2.00), and Al-Fajr (2.10).

Meccan vs. Medinan surahs. Medinan surahs are systematically harder: mean 3.49 vs. 2.98 words (17% more). This likely reflects the higher density of shared legislative, theological, and community-oriented vocabulary in the Medinan corpus.

What predicts difficulty? The strongest predictor is not ayah length but *vocabulary uniqueness*. The correlation between a surah’s hapax density (fraction of words appearing only once in the Quran) and its mean disambiguation length is $r = -0.54$ —a strong negative relationship. Surahs with more unique vocabulary are substantially easier. Ayah length, by contrast, correlates only weakly ($r = 0.11$).

Surah heatmaps. Figure 5 shows the disambiguation “fingerprint” of Surat al-Rahman (Q 55). The periodic vertical bands correspond to its 31-fold refrain. Narrative surahs like Yusuf (Q 12) show uniformly low difficulty (see Appendix E).

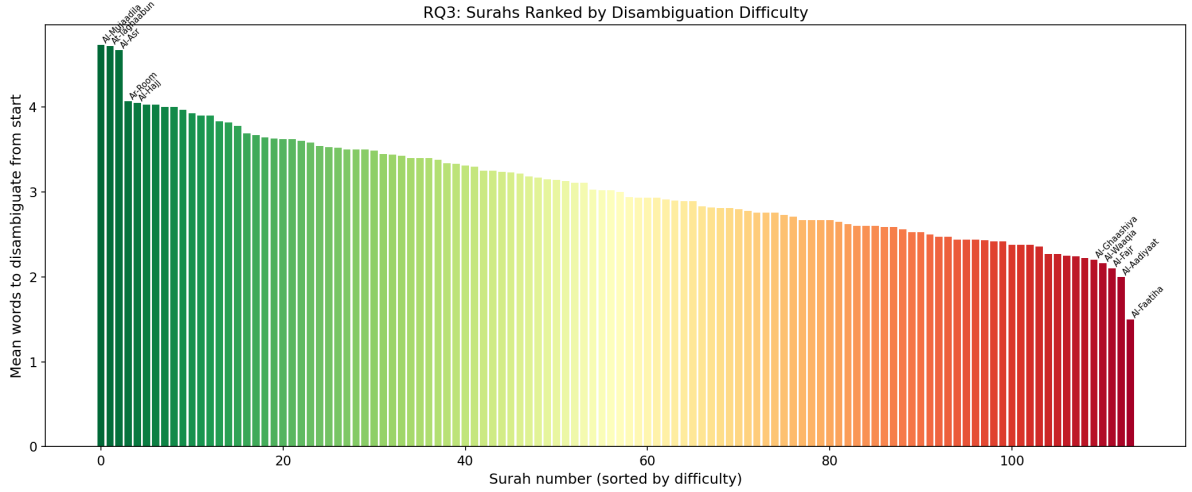


Figure 4: All 114 surahs ranked by mean disambiguation length.

Ayah A	Ayah B	Shared	Cross
Q 4:43	Q 5:6	24	✓
Q 11:110	Q 41:45	18	✓
Q 4:116	Q 4:48	17	
Q 13:25	Q 2:27	17	✓
Q 34:2	Q 57:4	16	✓
Q 2:134	Q 2:141	15	
Q 59:1	Q 61:1	15	✓

Table 2: Top confuser pairs by shared word-sequence length.

By Juz. *Huffāz* organize review by *juz*’ (one of 30 equal parts). The easiest is the 27th (mean 2.52), containing many short Meccan surahs. The most difficult is the 28th (mean 3.88). Later *ajza* (parts 16–30) are easier overall than earlier ones (mean 3.09 vs. 3.42).

3.5 Confuser Pairs and Persistent Confusers

We extract 250,482 pairs of ayahs sharing at least one consecutive word. The longest shared sequence spans **24 words** between Q 4:43 and Q 5:6—both addressing the rulings of ablution and dry purification.

At ≥ 5 shared words, 69.2% of confuser pairs are cross-surah—a wrong match leads to a completely different location in the text.

Persistent confusers. In the trace of Q 2:14, we saw Q 2:76 hold on as the single remaining confuser for 6 consecutive words. We call the last confuser standing in any narrowing curve the *persistent confuser* of that ayah. 886 ayahs (14.2%) have this “cliff-then-plateau” pattern: candidates drop rapidly to a single persistent confuser that holds on for two or more additional words before the ayahs finally diverge.

These relationships are often asymmetric. Q 34:2 requires 17 words to distinguish itself from Q 57:4, but Q 57:4 needs only 4 words to distinguish itself from Q 34:2—and its own persistent confuser is Q 2:29, not Q 34:2. 677 pairs are *symmetric*: each considers the other its persistent confuser. These are the true twins of the Quran—pairs locked together until the very last distinguishing word.

Confuser graph. A graph with edges for pairs sharing ≥ 5 words yields 1,821 nodes and 2,805 edges across 429 connected components. The largest spans 605 ayahs (Figure 6). The highest-degree hub is

Q2:255 (*Āyat al-Kursī*, 38 connections)—a long, doctrinally rich ayah. Despite containing 10 of the top 100 repeated phrases, it remains identifiable in 8 words from the start. Its distinctiveness comes not from unique vocabulary but from the unique *arrangement* of shared building blocks.

3.6 Repeated Phrases

1,957 word sequences appear in five or more distinct ayahs. The most common: **إِنَّ اللَّهَ** (“Indeed, God...”) in 243 ayahs, **الَّذِينَ ءَامَنُوا** (“those who believe”) in 182, and **فِي الْأَرْضِ** (“in the earth”) in 170.

The phrases with the highest disambiguation cost are *closing phrases* that appear at the end of many ayahs—because no words follow them, they can never resolve on their own. **غَفُورٌ رَحِيمٌ** (“Most Forgiving, Most Merciful”) closes 42 ayahs; **رَبِّ الْعَالَمِينَ** (“Lord of the Worlds”) closes 34.

Some ayahs are composed almost entirely of shared building blocks. Q 64:1 and Q 49:1 achieve 100% coverage by the top 100 repeated phrases—virtually every word appears as part of a recurring phrase elsewhere. At the other extreme, surahs like Quraysh (45% hapax density) and Al-Duha (38.5%) are built from highly distinctive vocabulary.

3.7 Continuous Recitation: Cross-Boundary Analysis

The analyses above treat each ayah in isolation. In practice, recitation is continuous. We construct the entire Quran as a single 78,092-word stream and analyze disambiguation at each of the 6,235 ayah boundaries.

For each boundary, we form *crossing windows*: the last t words of the preceding ayah concatenated with the first h words of the next. We check whether this crossing sequence appears at exactly one position in the continuous stream.

The results (Figure 7): 99.2% of boundaries are identifiable within 10 words per side, with a mean of 2.22 words (median 2). Only 52 boundaries remain ambiguous.

The most important finding: of the 339 ayahs that are never unique in isolation, **328 (96.8%) become identifiable** with boundary context. 269 of those need just 2 crossing words. 52 need 3–4. Only 7 need 5 or more.

Here are examples of how boundary context resolves otherwise-identical ayahs:

The basmalah. Q 1:1 (**بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ**) has 113 confusers—it opens nearly every surah. But the ayah *after* each basmalah is different: Q 1:1 is followed by Q 1:2 (**الْحَمْدُ لِلَّهِ رَبِّ الْعَالَمِينَ**), while Q 2’s basmalah is followed by Q 2:2 (**ذَلِكَ الْكِتَابُ**). One word across the boundary is enough.

Al-Rahman’s refrain. All 31 instances of **فَيَأْتِي ءَالَآءُ رَبِّكَمَا تُكَذِّبَانِ** are identical—but each is preceded and followed by a different ayah. Q 55:13 is preceded by Q 55:12 (about the two types of grain) and followed by Q 55:14 (creation from clay). Q 55:16 is preceded by Q 55:15 (creation from fire). Just 2 crossing words resolve all 31 instances.

Verbatim twins. Q 2:134 and Q 2:141 share all 15 words (تِلْكَ أُمَّةٌ قَدْ خَلَتْ). But Q 2:134 is preceded by Q 2:133 (about the testament of Ya‘qub, peace be upon him) and Q 2:141 is preceded by Q 2:140 (about claiming prophets’ religious affiliation). Two crossing words resolve the pair.

3.8 The 11 Truly Unresolvable Ayahs

Only 11 ayahs in the entire Quran remain unidentifiable even with boundary context. What makes them special is not just that they are repeated—but that their *neighbors* are also repeated in the same pattern. The ayah, the ayah before it, and the ayah after it all appear identically at another location, creating a sequence of three or more consecutive shared ayahs.

Here is the complete list, traced to show why boundary context fails:

Algorithm 5 Q 26:108 — “So be mindful of God and obey me” (also Q 26:126, 26:144, 26:163, 26:179)

Position	Ayah	Status
Before	Q 26:107: إِنِّي لَكُمْ رَسُولٌ أَمِينٌ	Also repeated 5×
This	Q 26:108: فَاقْبَلُوا آلِهَةً وَأَطِيعُوا	Repeated 5×
After	Q 26:109: وَمَا أَسْأَلُكُمْ عَلَيْهِ مِنْ أَجْرٍ	Also repeated 5×

In Surat al-Shu‘ara’ (Q 26), five prophets each deliver the same three-ayah sequence: “I am a trustworthy messenger to you / So be mindful of God and obey me / I ask no reward from you. . .” This entire block—before, during, and after—repeats identically five times (at ayahs 107–109, 125–127, 143–145, 162–164, 178–180). No amount of local context can distinguish which prophet’s story is being recited.

Algorithm 6 Q 15:37 and Q 38:80 — “He said: You are among those given respite”

Position	Q 15:37	Q 38:80
Before	Q 15:36: قَالَ رَبِّ فَأَنْظِرْنِي	Q 38:79: same text
This	Q 15:37: قَالَ فَإِنَّكَ مِنَ الْمُنْظَرِينَ	same text
After	Q 15:38: إِلَى يَوْمِ الْوَقْتِ الْمَعْلُومِ	Q 38:81: same text

The exchange between God and Iblis about being granted respite appears in both Surat al-Hijr and Surat Sad with identical wording across three consecutive ayahs.

Algorithm 7 Q 23:6–7 and Q 70:30–31 — Two consecutive shared ayahs

Position	Q 23:6–7	Q 70:30–31
Before	Q 23:5: same as Q 70:29	Q 70:29: same as Q 23:5
23:6/70:30	إِلَّا عَلَىٰ أَرْوَاحِهِمْ	same text (10 words)
23:7/70:31	فَمَنْ آتَبَعْنِي وَرَاءَ ذَلِكَ	same text (7 words)
After	Q 23:8: same as Q 70:32	Q 70:32: same as Q 23:8

Surat al-Mu‘minun and Surat al-Ma‘arij both contain a passage about the believers’ qualities, with four consecutive ayahs shared verbatim between the two surahs (Q 23:5–8 ↔ Q 70:29–32).

The pattern is consistent: all 11 unresolvable ayahs are embedded in *runs* of consecutive shared ayahs—sequences where not just the ayah itself but its entire local neighborhood is duplicated elsewhere. Five are from Al-Shu‘ara’ (the prophet-narrative refrain), two from the Iblis dialogue (Al-Hijr/Sad), two from Al-Mu‘minun, and two from Al-Ma‘arij.

To resolve these, a system would need context extending *beyond* the immediate neighbors—essentially knowing which surah is currently being recited. In continuous recitation, this context is always available from earlier in the recitation.

4 Structural Indexing

Minimum index depth. An index of all 25-grams suffices to uniquely fingerprint every ayah from at least one starting position.

Identifiability frontier. Figure 9 shows the fraction of ayahs identifiable as a function of n -gram size. At $n=1$: 9.5%. At $n=3$: 68.6%. At $n=6$: 89.4%. The curve plateaus near 94.6%.

Fingerprint set. The shortest unique identifying substring per ayah has a mean of 1.37 words (median 1). 39.4% of ayahs achieve their shortest fingerprint from a non-initial position—a prefix-only index would miss these.

5 What This Data Enables

We release this dataset in the hope that others will find uses for it we have not considered. Some directions we think are worth exploring:

Phonemic disambiguation. Our analysis is purely orthographic. Two ayahs that differ in spelling can sound nearly identical under *tajwīd* rules, and similar-looking ayahs may sound quite different. A *tajwīd*-aware phonemic layer—mapping text to IPA sequences—would capture a complementary dimension. We believe this is the highest-value extension.

Jump recovery. When a system commits to the wrong ayah at a shared phrase, how many additional words before the error becomes detectable? Our data could support such analysis, but the combinatorial space is large.

ASR confidence. The disambiguation map provides, at any word position, the number of candidate ayahs still in play. This could serve as a confidence signal for speech recognition systems.

Qira’at sensitivity. Our analysis uses the Hafs reading. How does the map change across the seven canonical readings?

Correlation with tafsir. Do the confuser clusters align with thematic groupings recognized in classical *tafsīr*? The 11-ayah clique around cosmological declarations suggests such connections may exist.

Adaptive systems. Could a recitation-assistance system use the narrowing curves to provide real-time feedback—waiting to commit until enough words have been heard, and indicating when a high-ambiguity region begins?

6 Limitations

This analysis operates on normalized orthographic forms only. It does not model phonemic similarity, morphological variation, or contextual meaning. Whitespace tokenization treats Arabic proclitics as part of their attached token; stem-aware matching changes disambiguation for approximately 26% of windows, with a modest overall increase (mean 2.74 to 2.85), confirming robustness while highlighting proclitic-aware tokenization as a refinement.

The cross-boundary model uses a fixed 10-word context window. A system mid-stream has the entire prior context, making our analysis conservative.

We make no claims about the memorization tradition. The Quran has been memorized by millions of people across 14 centuries, and the challenge of similar passages has been addressed by scholars far more qualified than us. This is a computational tool, nothing more.

7 Conclusion

We present a complete disambiguation map of the Quran: for every ayah, at every word position, how many words until identification is unique. The method is parameter-free, deterministic, and runs in 6.5 seconds. The median ayah is identifiable within 3 words. Under continuous recitation, 96.8% of never-unique ayahs become resolvable, leaving only 11 truly unresolvable ayahs—all embedded in runs of consecutive shared passages. The strongest predictor of a surah’s disambiguation difficulty is its proportion of unique vocabulary ($r = -0.54$), not its length.

We release six JSON artifacts (77 MB) under CC BY 4.0, along with all source code. We hope this resource proves useful, and that it serves, however modestly, the broader effort to develop respectful, high-quality tools for engaging with the Quran.

Wa Allāhu a‘lam (And God knows best).

References

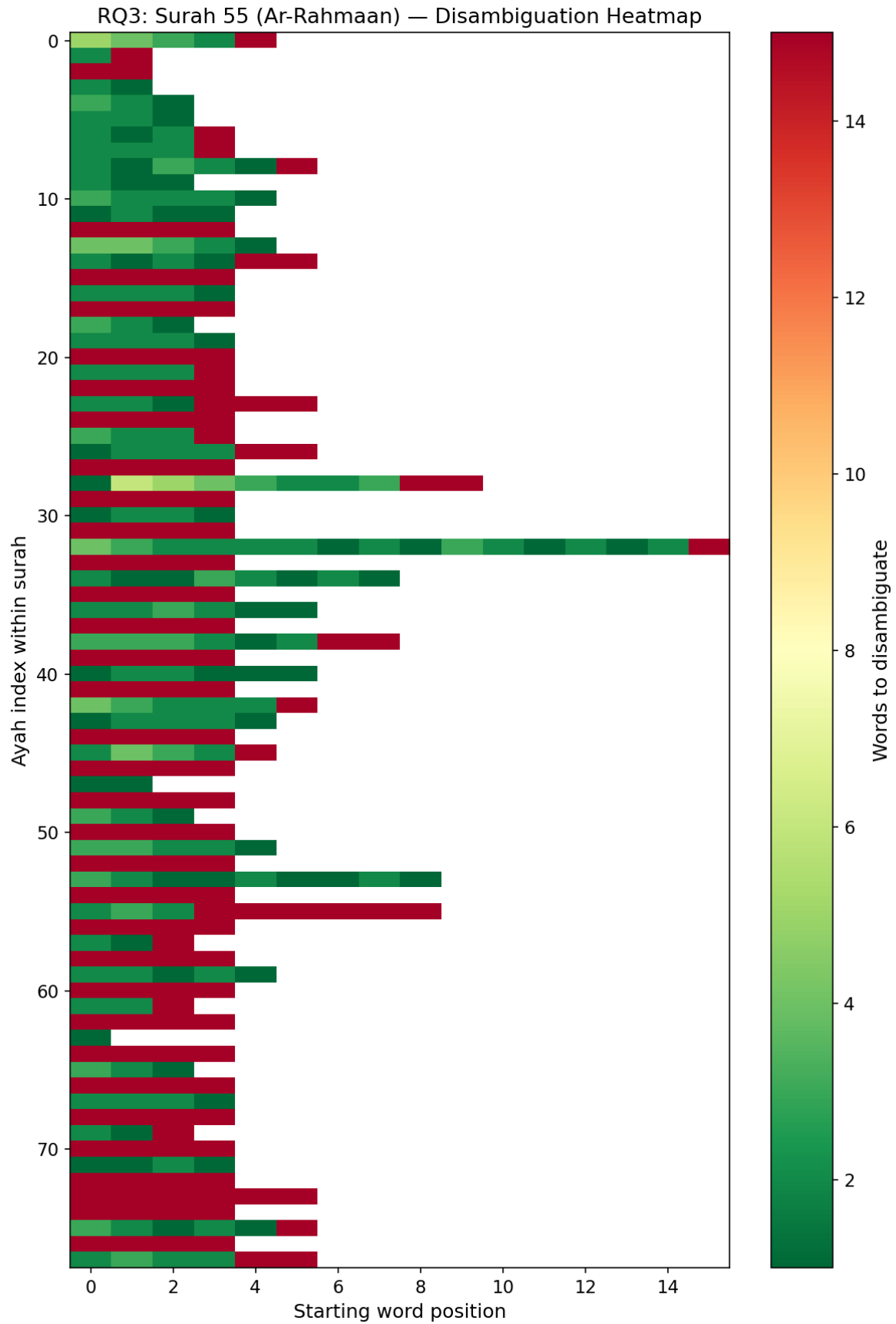


Figure 5: Surat al-Rahman (Q 55). Each row is an ayah; each column is a starting position. The vertical bands are the 31-fold refrain.

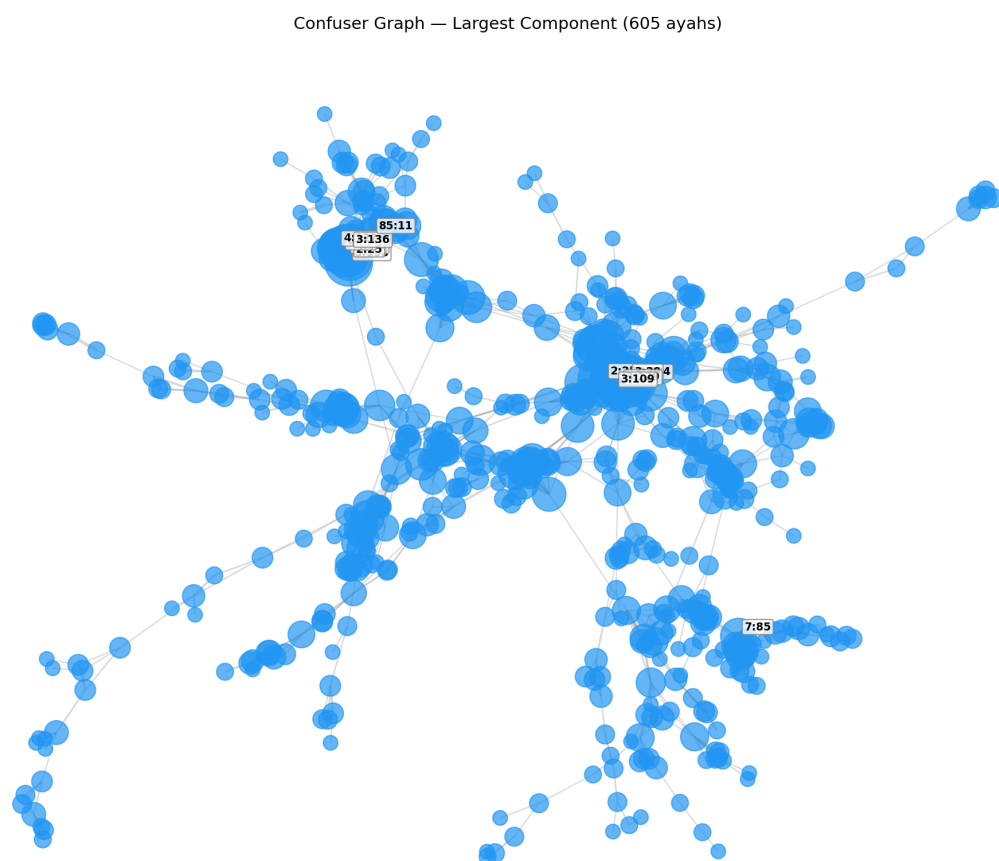


Figure 6: Largest connected component of the confuser graph (605 ayahs).

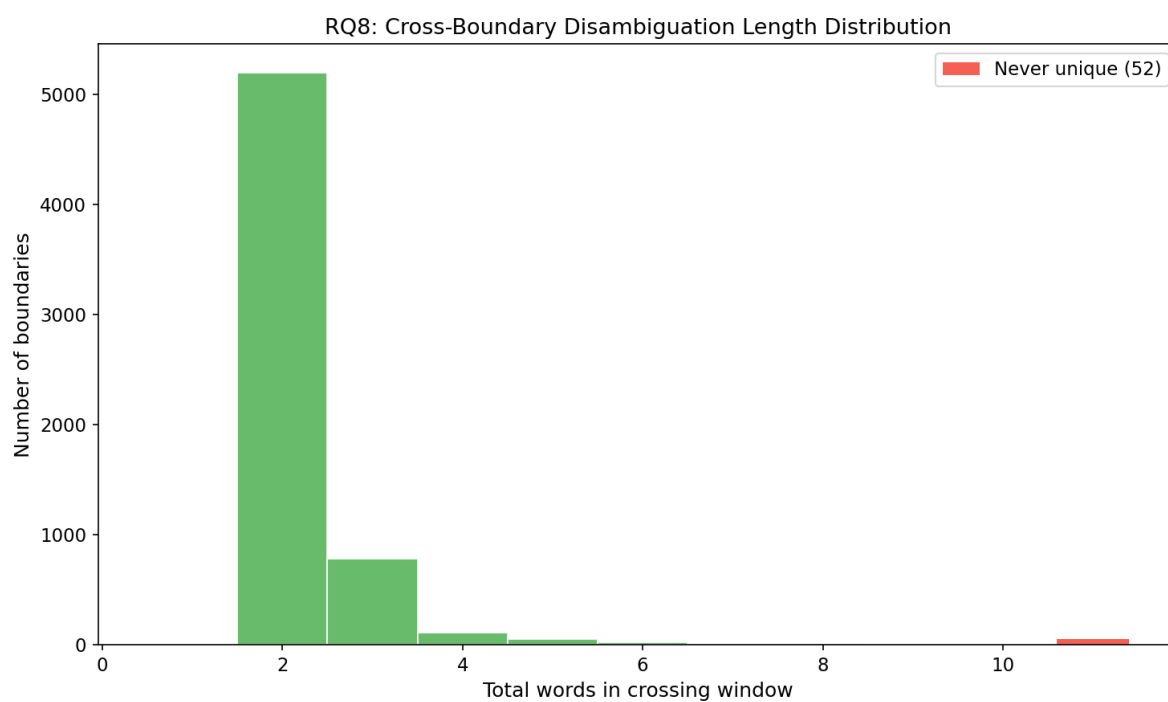


Figure 7: Crossing-window lengths at ayah boundaries. Most resolve with just 2 words.

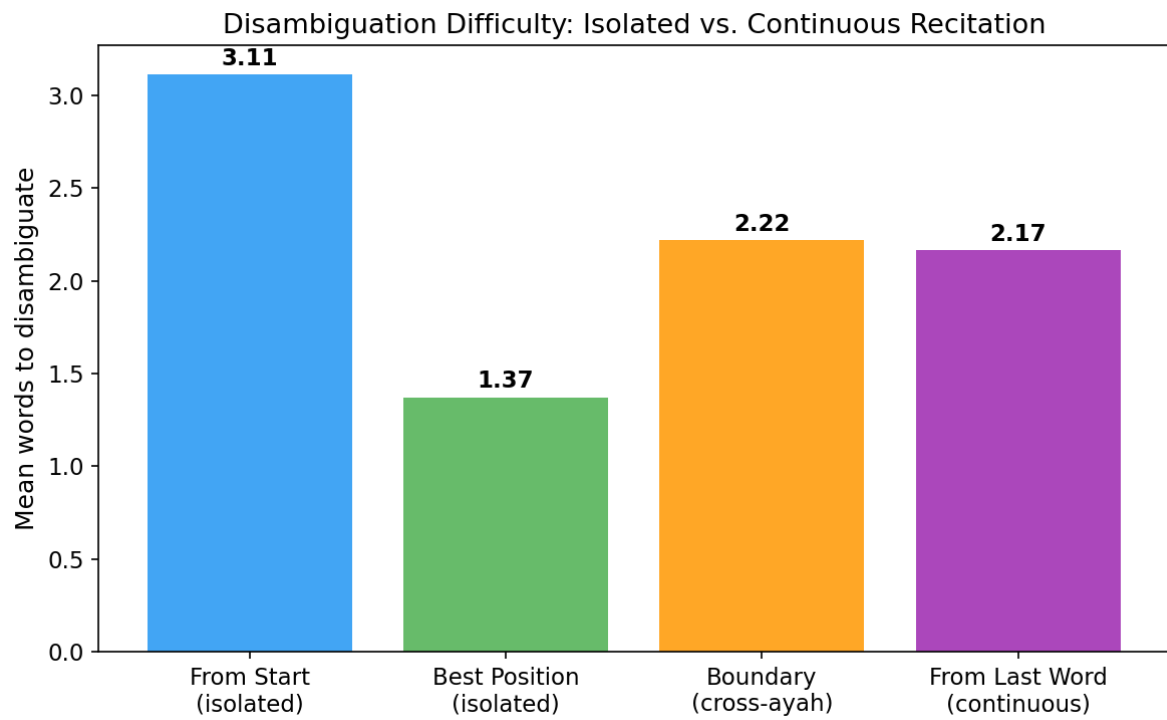


Figure 8: Mean disambiguation across four modes.

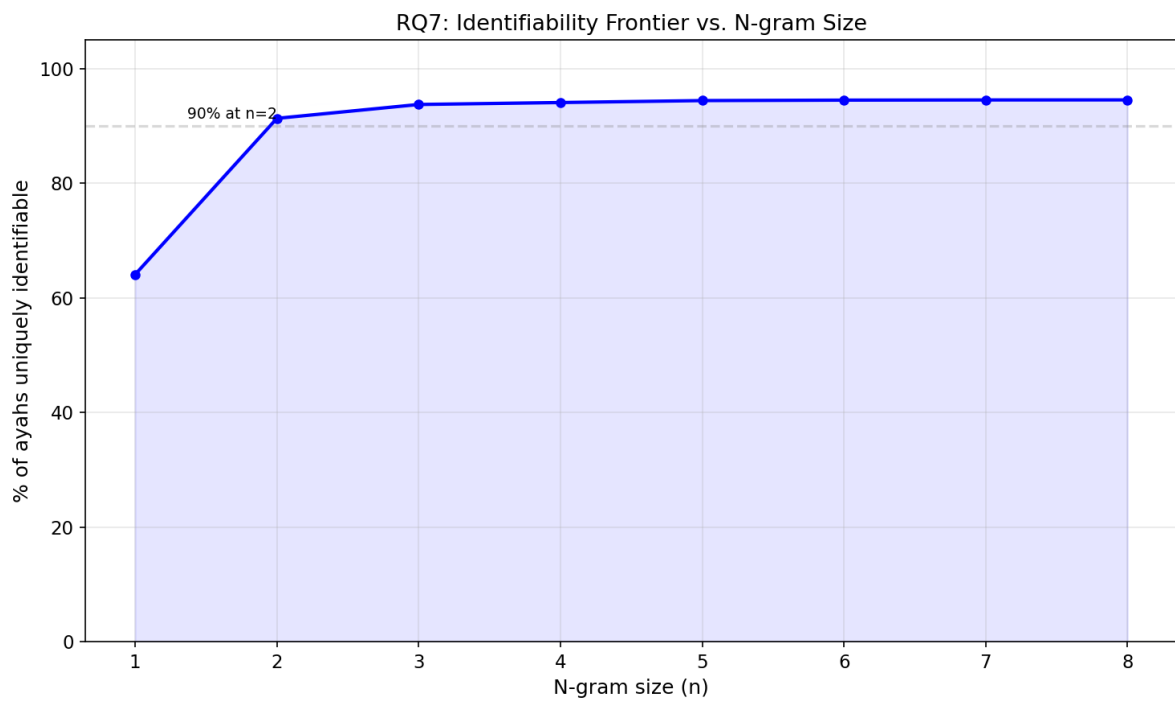


Figure 9: Fraction of ayahs identifiable vs. n -gram size.

A Dataset Artifacts

File	Size	Description
	69 MB	Full per-ayah, per-position narrowing curves
	619 KB	Lightweight: $d[]$ per ayah
	591 KB	Shortest unique substring per ayah
	32 KB	Top 500 pairs by shared length
	15 KB	200 most-repeated word sequences
	2.5 MB	Cross-boundary disambiguation

Table 3: Released dataset artifacts.

B Never-Unique Ayah Categories

Category	Count	Example
Basmalah	113+	Q 1:1
Al-Raḥmān refrain	31	Q 55:13–77
Al-Shu‘arā’ shared phrases	53	Q 26:8–191
Al-Şāffāt shared phrases	28	Q 37:40–182
Al-Mursalāt refrain	14	Q 77:15–50
Al-Ḥijr shared phrases	14	Q 15:29–93
Şād shared phrases	10	Q 38:3–87
Ḥurūf muqatta‘āt	8	Q 2:1
Other verbatim repeats	58	Q 2:134/141

Table 4: The 339 never-unique ayahs by category.

C Never-Unique Boundary Categories

#	Surah	Count	Cause
26	Al-Shu‘arā’	20	Prophet-narrative refrain
37	Al-Şāffāt	4	“Salāmūn ‘alā” phrases
15	Al-Ḥijr	4	“Innā la-naḥnu” refrain
23	Al-Mu‘minūn	3	Shared passage
38	Şād	3	Prophet-narrative echoes
70	Al-Ma‘ārij	3	Shared passage
<i>11 other surahs</i>		15	1–2 each

Table 5: The 52 never-unique boundaries by surah.

D All 114 Surahs

E Surah Heatmaps

F Top 10 Shared Closing Phrases

G Confuser Graph Statistics

H Reproducibility

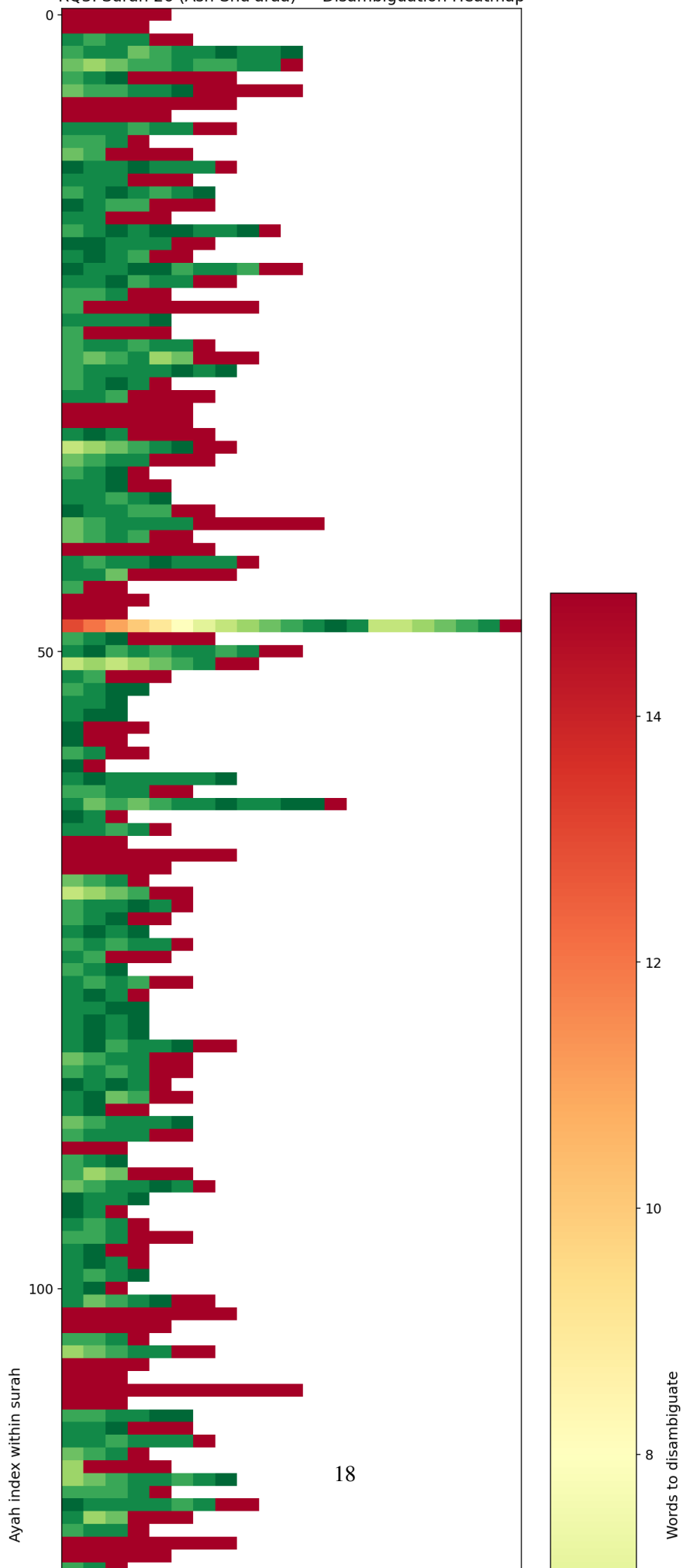
The analysis requires Python 3.11+ with no external dependencies beyond the standard library. The full pipeline completes in 6.5 seconds. Given identical input, the output is **bit-for-bit identical** across runs and platforms. All code and data: .

# Surah	Ay.	Mean	Med.	NU	Best	Hardest	<i>d</i>
1 Al-Fātiḥah	7	1.50	1.5	3	1.25	1:4	2
2 Al-Baqarah	286	3.78	3.0	8	1.35	2:48	12
3 Āl-i-‘Imrān	200	3.34	3.0	5	1.36	3:10	14
4 Al-Nisā’	176	3.52	3.0	0	1.28	4:48	18
5 Al-Mā’idah	120	3.64	3.0	2	1.27	5:17	11
6 Al-An‘ām	165	3.63	3.0	3	1.35	6:152	13
7 Al-A‘rāf	206	3.44	3.0	7	1.35	7:54	15
8 Al-Anfāl	75	3.43	3.0	0	1.32	8:39	8
9 Al-Tawbah	129	2.76	2.0	2	1.21	9:11	7
10 Yūnus	109	3.60	3.0	2	1.60	10:3	15
11 Hūd	123	3.49	3.0	2	1.30	11:50	14
12 Yūsuf	111	2.93	3.0	0	1.19	12:102	11
13 Al-Ra‘d	43	3.53	3.0	0	1.35	13:7	10
14 Ibrāhīm	52	3.02	3.0	1	1.25	14:2	10
15 Al-Ḥijr	99	3.13	3.0	14	1.52	15:19	10
16 Al-Naḥl	128	3.40	3.0	2	1.58	16:115	10
17 Al-Isrā’	111	3.25	3.0	1	1.22	17:34	13
18 Al-Kahf	110	2.91	2.0	2	1.15	18:110	12
19 Maryam	98	3.02	2.0	1	1.26	19:10	11
20 Ṭā Hā	135	2.80	2.0	2	1.32	20:71	13
21 Al-Anbiyā’	112	2.81	3.0	2	1.39	21:28	8
22 Al-Ḥajj	78	4.05	3.0	1	1.36	22:14	13
23 Al-Mu‘minūn	118	3.11	3.0	9	1.58	23:21	9
24 Al-Nūr	64	3.38	3.0	1	1.32	24:61	13
25 Al-Furqān	77	3.17	3.0	1	1.25	25:59	14
26 Al-Shu‘arā’	227	2.67	2.0	53	1.52	26:49	13
27 Al-Naml	93	2.93	3.0	3	1.40	27:69	9
28 Al-Qaṣaṣ	88	2.89	3.0	4	1.32	28:48	7
29 Al-‘Ankabūt	69	3.69	3.0	2	1.37	29:68	10
30 Al-Rūm	60	4.07	4.0	2	1.48	30:9	12
31 Luqmān	34	3.93	3.0	4	1.50	31:21	12
32 Al-Sajdah	30	3.14	3.0	1	1.59	32:22	7
33 Al-Aḥzāb	73	2.94	3.0	1	1.32	33:62	8
34 Saba’	54	3.58	3.0	1	1.36	34:2	17
35 Fāṭir	45	3.90	3.0	3	1.33	35:13	14
36 Yā Sīn	83	2.76	3.0	3	1.44	36:29	8
37 Al-Šaffāt	182	2.38	2.0	28	1.58	37:53	7
38 Šād	88	2.82	3.0	10	1.26	38:3	7
39 Al-Zumar	75	3.45	3.0	2	1.41	39:27	10
40 Ghāfir	85	3.22	3.0	3	1.48	40:82	12
41 Fuṣṣilat	54	3.50	3.0	2	1.58	41:6	12
42 Al-Shūrā	53	3.15	3.0	1	1.35	42:4	8
43 Al-Zukhruf	89	2.65	2.0	3	1.45	43:9	8
44 Al-Dukhān	59	2.71	2.0	4	1.60	44:8	7
45 Al-Jāthiyah	37	3.23	3.0	2	1.54	45:15	8
46 Al-Aḥqāf	35	4.03	3.0	2	1.39	46:7	9
47 Muḥammad	38	3.97	3.5	0	1.26	47:12	13
48 Al-Faṭḥ	29	3.54	3.0	1	1.43	48:17	13
49 Al-Hujurāt	18	3.83	3.0	0	1.22	49:1	9
50 Qāf	45	3.31	3.0	0	1.47	50:39	12
51 Al-Dhāriyāt	60	2.47	2.0	2	1.47	51:25	7
52 Al-Ṭūr	49	2.67	2.0	4	1.38	52:1	5
53 Al-Najm	62	2.42	2.0	0	1.35	53:31	8
54 Al-Qamar	55	2.47	2.0	8	1.30	54:1	5
55 Al-Raḥmān	78	2.22	2.0	33	1.22	55:1	5
56 Al-Wāqī‘ah	96	2.16	2.0	10	1.33	56:1	6
57 Al-Ḥadīd	29	4.03	4.0	0	1.62	57:1	10

# Surah	Ay.	Mean	Med.	NU	Best	Hardest	<i>d</i>
58 Al-Mujādilah	22	4.73	5.0	0	1.27	58:17	10
59 Al-Ḥashr	24	3.30	2.0	1	1.22	59:22	8
60 Al-Mumtaḥanah	13	3.62	3.0	0	1.23	60:1	9
61 Al-Šaff	14	4.00	4.0	2	1.33	61:7	7
62 Al-Jumu‘ah	11	3.90	3.0	1	1.50	62:1	13
63 Al-Munāfiqūn	11	3.82	3.0	0	1.18	63:2	8
64 Al-Taghābun	18	4.72	5.0	0	1.72	64:1	13
65 Al-Ṭālāq	12	3.00	2.5	0	1.08	65:1	7
66 Al-Tahrīm	12	3.18	3.0	1	1.09	66:1	7
67 Al-Mulk	30	3.03	3.0	1	1.38	67:1	5
68 Al-Qalam	52	2.38	2.0	7	1.36	68:1	5
69 Al-Ḥāqqah	52	2.44	2.0	7	1.31	69:19	6
70 Al-Ma‘ārij	44	2.59	2.0	7	1.51	70:1	5
71 Nūḥ	28	2.78	2.0	1	1.26	71:1	6
72 Al-Jinn	28	2.89	3.0	0	1.29	72:1	6
73 Al-Muzzammil	20	2.53	2.0	1	1.26	73:1	6
74 Al-Muddaththir	56	2.27	2.0	1	1.36	74:1	6
75 Al-Qiyāmah	40	2.44	2.0	1	1.31	75:1	7
76 Al-Insān	31	2.93	3.0	1	1.20	76:30	7
77 Al-Mursalāt	50	2.44	2.0	14	1.33	77:1	5
78 Al-Naba’	40	2.67	2.0	1	1.36	78:35	6
79 Al-Nāzi‘āt	46	2.24	2.0	4	1.33	79:1	5
80 ‘Abasa	42	2.27	2.0	2	1.30	80:1	5
81 Al-Takwīr	29	2.81	2.0	2	1.30	81:29	7
82 Al-Infīṭār	19	2.83	2.0	1	1.56	82:1	7
83 Al-Muṭaffifīn	36	2.59	2.0	7	1.45	83:1	6
84 Al-Inshiqāq	25	2.90	3.0	4	1.67	84:1	7
85 Al-Burūj	22	3.24	3.0	1	1.52	85:11	8
86 Al-Ṭariq	17	2.76	3.0	0	1.12	86:1	6
87 Al-A‘lā	19	2.53	2.0	0	1.68	87:1	6
88 Al-Ghāshiyah	26	2.20	2.0	1	1.20	88:1	6
89 Al-Fajr	30	2.10	2.0	0	1.17	89:6	6
90 Al-Balad	20	2.60	2.0	0	1.45	90:1	7
91 Al-Shams	15	2.73	3.0	0	1.07	91:1	5
92 Al-Layl	21	2.38	2.0	0	1.62	92:1	5
93 Al-Ḍuḥā	11	2.36	2.0	0	1.09	93:1	5
94 Al-Sharḥ	8	2.50	2.0	0	1.38	94:1	6
95 Al-Tīn	8	3.25	2.5	0	1.25	95:6	6
96 Al-‘Alaq	19	2.42	2.0	0	1.47	96:1	5
97 Al-Qadr	5	3.40	3.0	0	2.20	97:1	6
98 Al-Bayyinah	8	3.62	3.5	0	1.50	98:7	6
99 Al-Zalzalah	8	2.62	2.0	0	1.50	99:1	6
100 Al-‘Ādiyāt	11	2.00	2.0	0	1.09	100:1	5
101 Al-Qārī‘ah	11	3.11	3.0	2	1.67	101:1	5
102 Al-Takāthur	8	2.43	2.0	1	1.57	102:1	5
103 Al-‘Aṣr	3	4.67	5.0	0	1.67	103:3	6
104 Al-Humazah	9	2.56	2.0	0	1.22	104:1	6
105 Al-Fil	5	2.60	2.0	0	1.00	105:1	6
106 Quraysh	4	2.25	1.5	0	1.00	106:1	5
107 Al-Mā‘ūn	7	2.67	2.5	1	1.67	107:1	5
108 Al-Kawthar	3	3.33	2.0	0	1.00	108:1	6
109 Al-Kāfirūn	6	3.50	3.0	2	1.75	109:1	6
110 Al-Naṣr	3	3.67	4.0	0	1.00	110:1	6
111 Al-Masad	5	2.60	2.0	0	1.20	111:1	5
112 Al-Ikhlāṣ	4	3.50	3.0	0	1.75	112:1	6
113 Al-Falaq	5	4.00	3.0	0	1.20	113:1	8
114 Al-Nās	6	3.40	2.0	1	1.60	114:1	8

Table 6: All 114 surahs. Mean/Med = disambiguation from start (excl. never-unique). NU = never-unique count. Best = mean best-position *d*. Hardest = most demanding ayah with its *d* value.

RQ3: Surah 26 (Ash-Shu'araa) — Disambiguation Heatmap



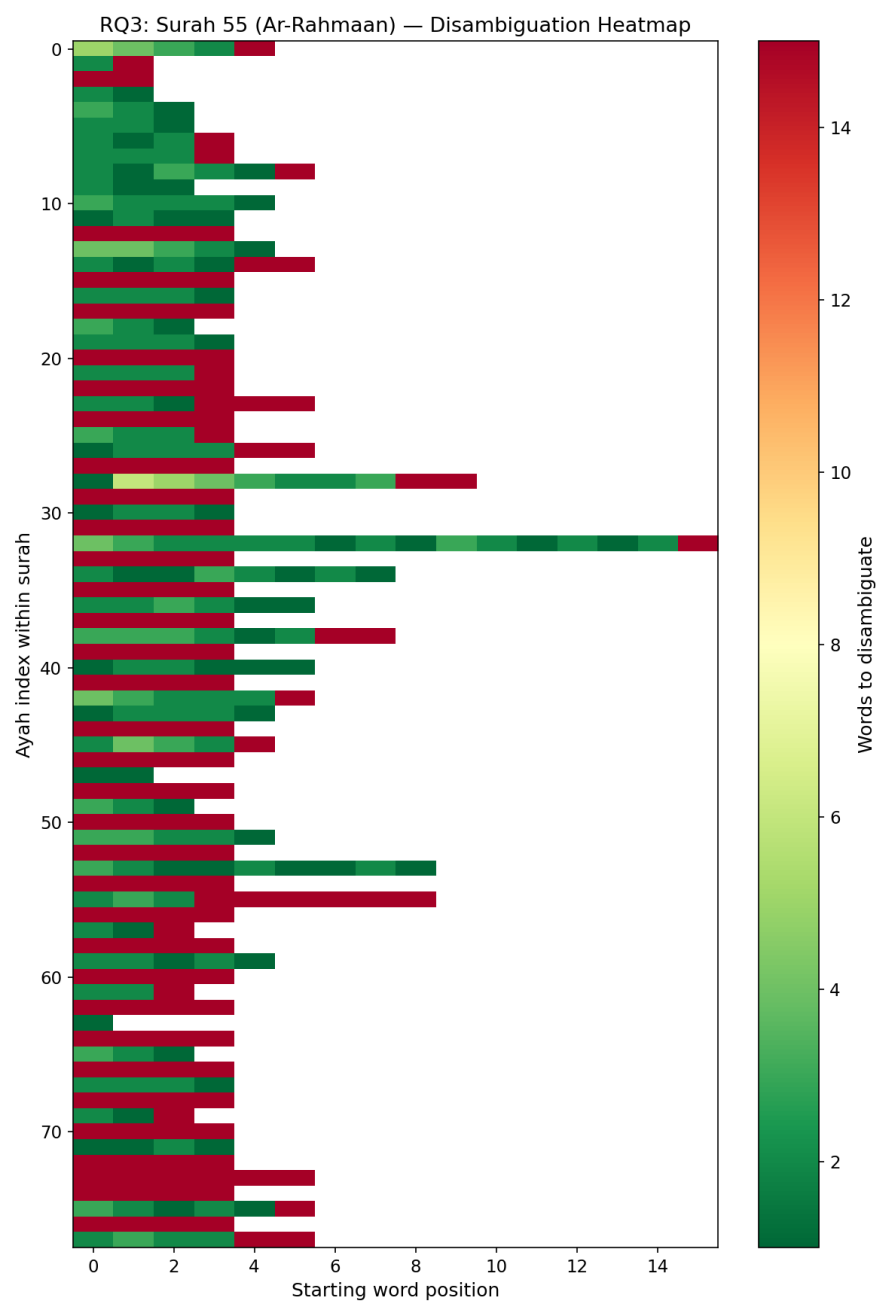


Figure 11: Surat al-Rahman (Q 55): vertical bands from the 31-fold refrain.

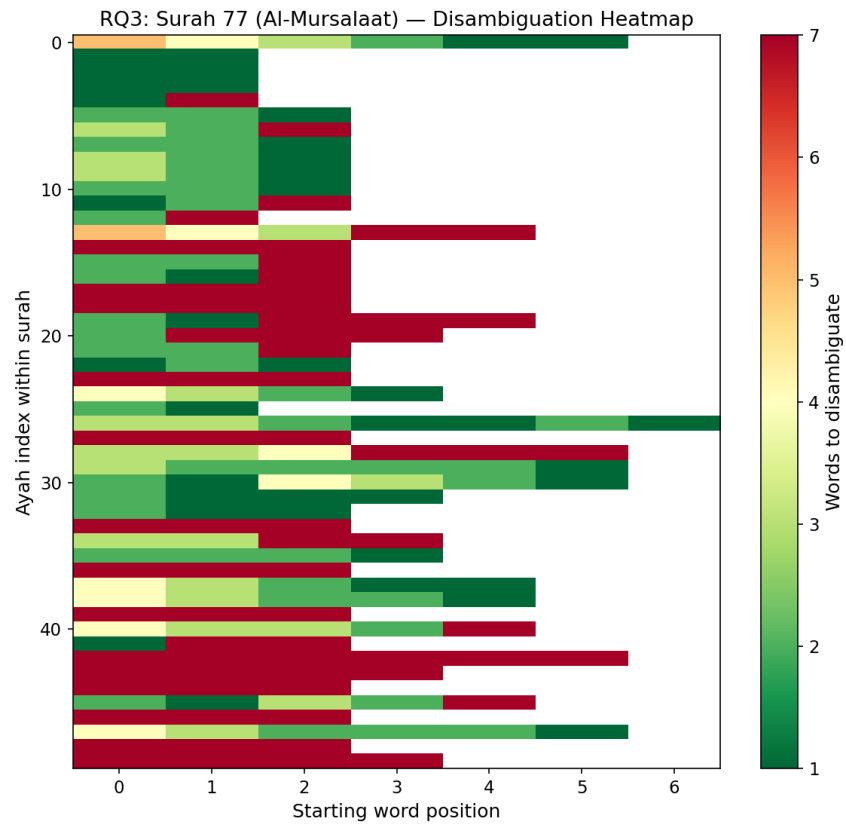


Figure 12: Surat al-Mursalat (Q 77): banding from 10-fold refrain.

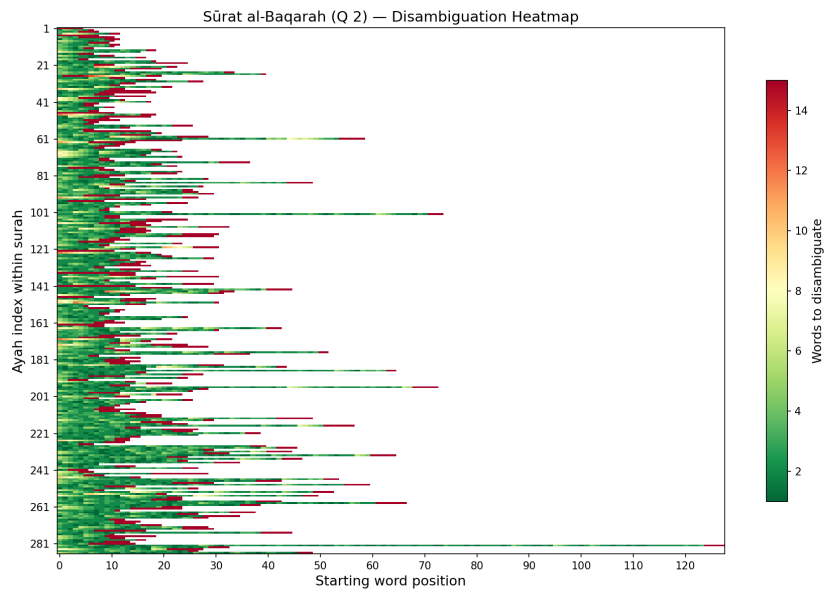


Figure 13: Surat al-Baqarah (Q 2): 286 ayahs with scattered peaks.

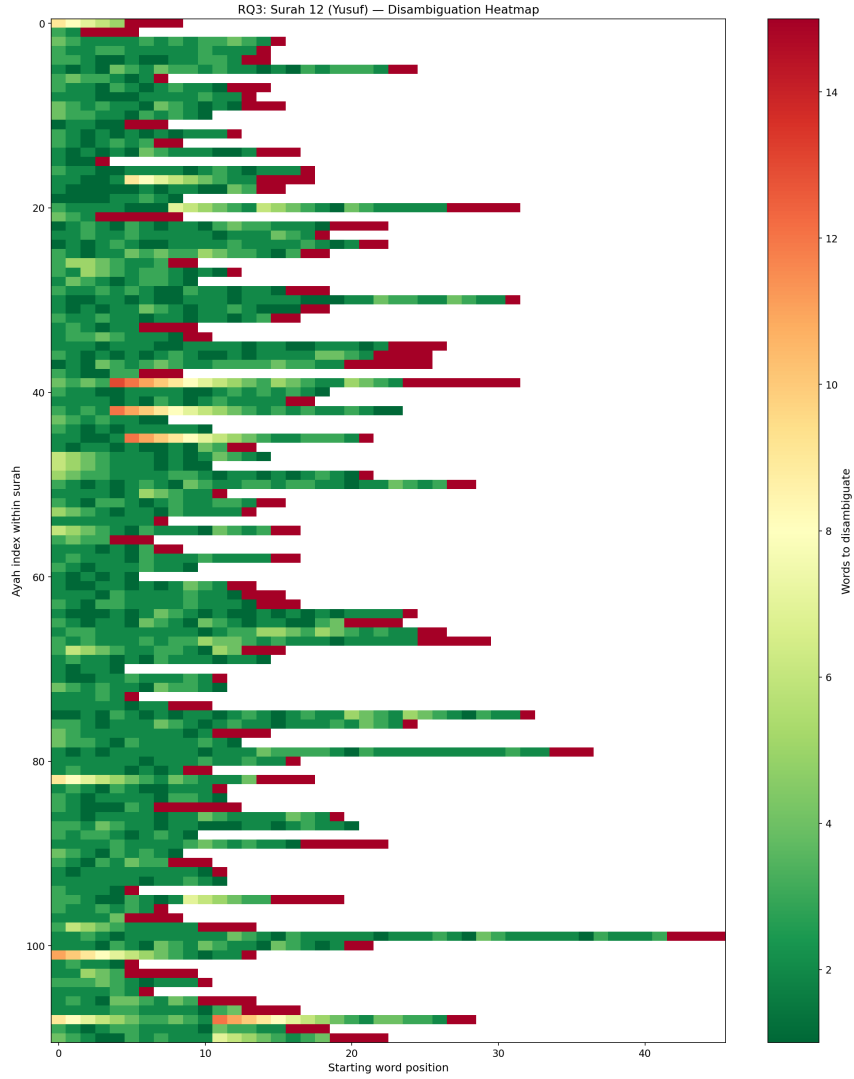


Figure 14: Surat Yusuf (Q 12): a narrative surah with uniformly low difficulty.

Phrase	Freq.	NUR
<i>ghafūrun raḥīm</i> (Most Forgiving, Most Merciful)	42	100%
<i>rabbu al-‘ālamīn</i> (Lord of the Worlds)	34	100%
<i>fa-bi-ayyi ālā’i...</i> (Which of your Lord’s favors...)	31	100%
<i>al-‘azīzu al-ḥakīm</i> (the Almighty, the All-Wise)	29	100%
<i>in kuntum ṣādiqīn</i> (if you are truthful)	28	100%
<i>allāhu ghafūrun raḥīm</i> (God is Most Forgiving, Most Merciful)	22	100%
<i>fīhā khālīdūn</i> (therein they abide)	19	100%
<i>al-qawm al-ẓālimīn</i> (the wrongdoing people)	19	100%
<i>mā kānū ya‘malūn</i> (what they used to do)	18	100%
<i>bi-kulli shay’in ‘alīm</i> (Knower of all things)	16	100%

Table 7: Top 10 shared closing phrases. NUR = never-unique rate.

Metric	Value
Nodes (≥ 5 -word overlap)	1,821
Edges	2,805
Connected components	429
Largest component	605 nodes
Max degree (Q 2:255)	38
Cross-surah edges	69.2%
Symmetric persistent-confuser pairs	677
Cliff-then-plateau ayahs	886 (14.2%)

Table 8: Confuser graph statistics (≥ 5 shared words).